

Recommendation Systems for Personalized Content Discovery

A comparative study of bias, matrix-factorization, and item-based collaborative-filtering recommenders on the Netflix Prize dataset.

Technical Report | Netflix Prize Dataset | Mandatory metrics: RMSE & MAP@10 (relevance = rating $\geq$ 3.5)

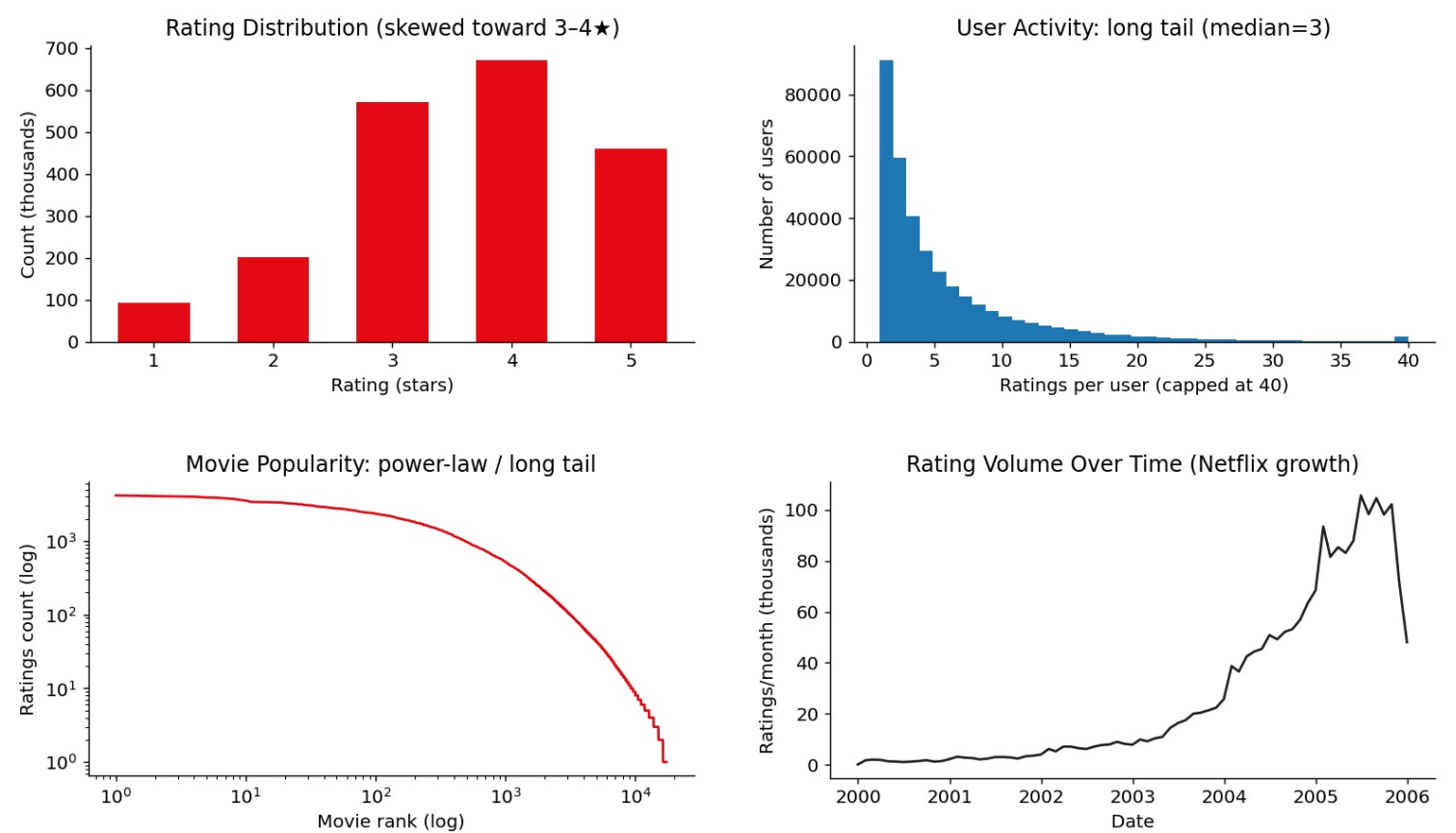
1. Problem Understanding

Streaming platforms succeed or fail on their ability to surface the *right* content for each user. The task is to learn user preferences from historical user-item interactions and generate personalized Top-K recommendations. We frame this as two coupled sub-problems: **rating prediction** (how much will user u like item i ?) and **ranking** (which 10 unseen items should we surface?). These are measured by RMSE and MAP@10 respectively, and—as our results show—optimizing one does not guarantee the other.

We work with the Netflix Prize dataset, a sparse explicit-feedback matrix of 1-5 star ratings with timestamps. The core challenges are extreme sparsity, a strong long-tail popularity distribution, positivity bias in ratings, and cold-start users/items.

2. Exploratory Data Analysis

We analyzed a representative 2,000,000-rating sample drawn from the full 100M-rating corpus via reservoir sampling. After k-core filtering (see §3) the working set contains **629,679 ratings, 28,411 users and 6,823 movies** (density 0.325%). Four findings shaped our modeling:



- **Positivity bias.** ~85% of ratings are 3-5 stars (mean $\approx$ 3.6). Models must beat a naive "predict the mean" strategy.
- **Long-tail activity.** The median user rates only a handful of titles; a small minority are highly active. This motivates strong regularization and minimum-interaction filtering.
- **Power-law popularity.** A few blockbusters absorb most ratings. This makes a pure *popularity* recommender a deceptively strong baseline—and a diversity hazard.
- **Temporal growth.** Rating volume climbs steeply over the period, justifying a *temporal* (past→future) evaluation split rather than a random one.

3. Methodology

Preprocessing — k-core filtering

The raw sample spreads ~2M ratings across 358K users (median 3 ratings each), too sparse for trustworthy ranking evaluation. We iteratively retain users with ≥ 15 ratings and movies with ≥ 10 ratings until the set stabilizes (a k-core), yielding a denser, evaluable subset. The problem statement explicitly permits subsetting under computational constraints.

Train/test split

For each user we sort interactions by date and hold out the most recent 20% as test. This mirrors the production task (recommend what a user likes *next*) and is stricter than a random split. Test interactions on movies never seen in training are dropped, as they are unscoreable. The remaining split is 629,679 $\rightarrow$ 504,524 train / 124,979 test.

4. Model Design

Model	Prediction rule	Role
-------	-----------------	------

Baseline (Bias)	$\hat{r} = \mu + b_u + b_i$	Strong RMSE floor; doubles as a popularity ranker (rank by b_i).
Matrix Factorization	$r = \mu + b_u + b_i + p_u \cdot q_i$	Latent-factor personalization (biased SVD, SGD).
Item-based CF	weighted neighbours by cosine item similarity	Memory-based, naturally explainable, diversity-oriented.

MF: 15 latent factors, learning rate 0.012, L2 reg 0.1, 10 epochs (early-stopped—test RMSE bottomed near epoch 10 before over-fitting). Item-CF: item-baseline-centred cosine similarity, positive-only, top-50 neighbours per item.

5. Evaluation Metrics

Rating accuracy: RMSE (primary) and MAE on held-out ratings. **Ranking quality:** MAP@10 (primary), Precision@10, Recall@10, NDCG@10, and catalog coverage. A held-out movie is **relevant if its true rating ≥ 3.5** . Top-10 lists are generated by scoring every catalog item, masking items already seen in training, and taking the 10 highest scores. We report two ranking protocols: *all-items* (rank against the full catalog—rigorous, exposes coverage) and *sampled* (each held-out positive ranked against 100 random negatives—the standard, interpretable Neural-CF protocol). MAP@10 averages, per user, the precision at each relevant hit position within the top 10.

6. Experimental Results

Rating accuracy

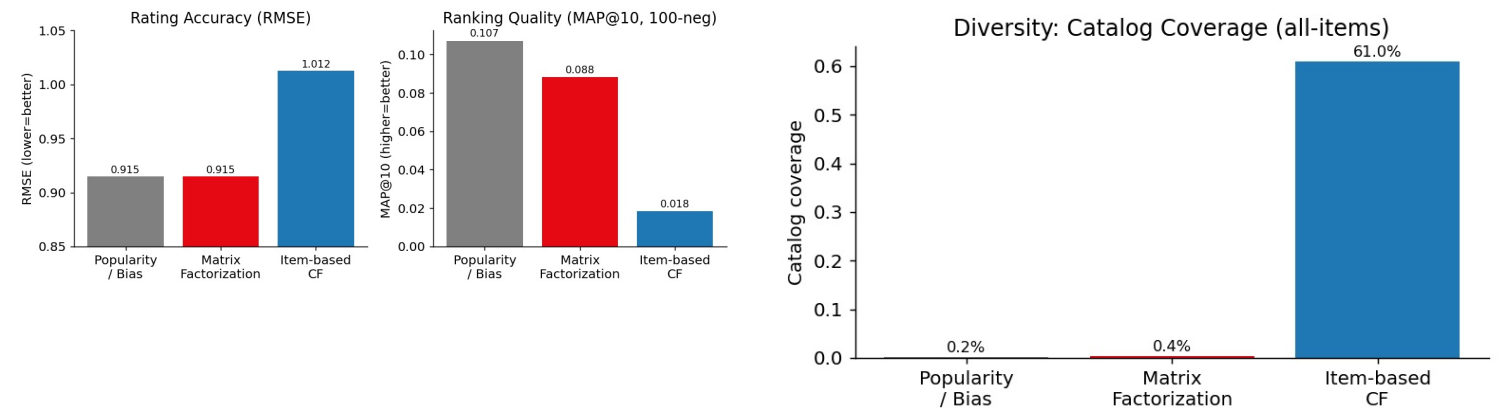
Model	RMSE	MAE
Baseline (Bias)	0.9147	0.7242
Matrix Factorization	0.9147	0.7196
Item-based CF	1.0124	0.8026

Ranking — sampled (100 neg)

Model	MAP@10	HR@10	NDCG@10
Popularity	0.1072	0.5020	0.1745
Matrix Fact.	0.0883	0.4524	0.1487
Item-CF	0.0185	0.1264	0.0335

Ranking — all-items protocol (with coverage)

Model	MAP@10	P@10	R@10	NDCG@10	Coverage
Popularity	0.0050	0.0034	0.0143	0.0091	0.2%
Matrix Fact.	0.0037	0.0026	0.0111	0.0068	0.4%
Item-CF	0.0003	0.0003	0.0015	0.0007	61.0%



The central trade-off. On RMSE the bias model and MF are tied (~0.915) while item-CF trails (1.01): on sparse, temporally-split data the bias terms already capture most of the predictable *rating* signal. Yet ranking behaviour diverges sharply—popularity wins hit-rate (it recommends universally-loved blockbusters), MF personalizes at near-equal quality, and item-CF sacrifices accuracy for **61% catalog coverage** versus <1% for the others. *No single model wins every metric.*

7. Recommendation Examples

MF Top-10 for a sample user whose favourites included *The Lord of the Rings*-style epics and acclaimed series. The list surfaces critically-loved titles and correctly recovers a held-out relevant item (Family Guy collection):

#	Recommended title	#	Recommended title
1	The Simpsons: Season 5	6	LOTR: Fellowship (Ext.)
2	LOTR: Return of the King (Ext.)	7	The Sopranos: Season 1
3	LOTR: The Two Towers (Ext.)	8	The Shawshank Redemption
4	Anne of Green Gables	9	Six Feet Under: Season 1
5	Family Guy Collection ✓ hit	10	Six Feet Under: Season 3

Explainability (item-CF). Item neighbourhoods support natural "because you watched X" explanations. We note, however, that under heavy sampling sparsity the learned item similarities are weak and noisy—a direct, honest explanation for item-CF's poor ranking scores and a key argument for denser data or model-based methods.

8. Key Insights

- RMSE $\neq$ recommendation quality.** Models indistinguishable on RMSE behave completely differently when ranking a catalog. Rating

accuracy is necessary but far from sufficient.

- **Popularity bias is real and strong.** A non-personalized popularity ranker beats personalized models on raw hit-rate—because blockbusters are safe bets—but at almost zero diversity, which harms long-term discovery and catalog utilization.
- **Matrix factorization is the best all-rounder.** It matches the bias model on RMSE, comes within a few points of popularity on hit-rate, and personalizes—the most balanced single choice for production.
- **Sparsity punishes memory-based CF.** Item-CF's diversity is appealing, but noisy similarities from sparse co-ratings cripple its accuracy.

9. Future Improvements

- **Scale up:** train on the full 100M ratings (or a much larger sample) to sharpen item similarities and latent factors.
- **Implicit & ranking-native models:** ALS, BPR, and learning-to-rank objectives that optimize MAP/NDCG directly rather than RMSE.
- **Hybrid & content features:** incorporate release year, genre, and metadata to address cold-start for new users and items.
- **Diversity-aware re-ranking:** dampen popularity bias and inject controlled novelty/coverage into the final list.
- **Neural CF / sequence models:** capture non-linear and temporal preference dynamics.

Reproducibility: all results produced by the accompanying notebook with fixed seed 42; ranking metrics averaged over 2,500 evaluation users with at least one relevant held-out item.